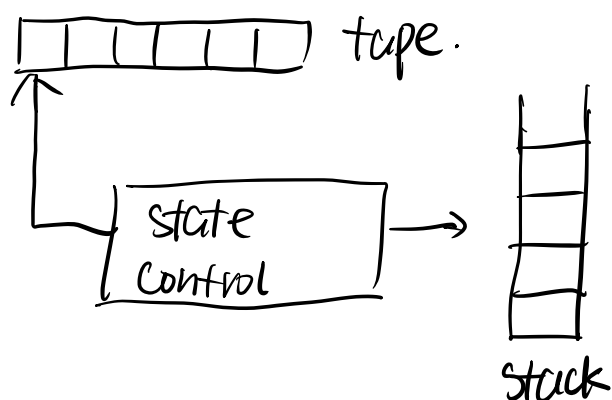


Pushdown Automata

PDA = NFA + stack



Definition

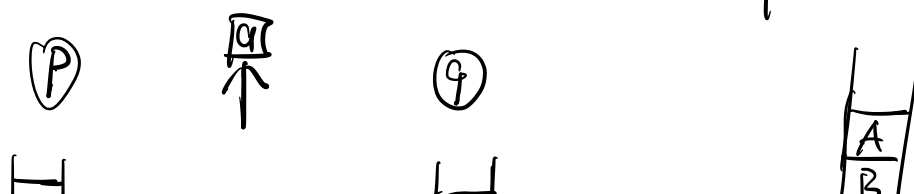
A pushdown automata (PDA) is a 6-tuple $P = (K, \Sigma, \Gamma, \Delta, s, \bar{F})$

- K is a set of states
- Σ is a set of input symbol
- Γ is a set of stack symbol
- $s \in K$ is the initial state
- $\bar{F} \subseteq K$ is the set of final state
- Δ , transition relation, is a *finite* subset of

$(K \times (\Sigma \cup \epsilon) \times \Gamma^*) \times (K \times \Gamma^*)$
 / | | | \
 current symbol a string next a string
 state on the state we will
 top of write into
 the stack the stack

$((p, a, \alpha), (q, \beta))$

$\beta = ABC$





$(p, a, e), (q, \beta)$

p

a

q



A configuration for P is a member of $K \times \Sigma^* \times \Gamma^*$

$\downarrow$ current state $\downarrow$ unread input $\downarrow$ stack content

$(q, a \circ, abcd)$



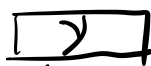
$(p, x, \alpha) \vdash_P (q, y, \beta)$ if $\exists (p, a, r), (q, \eta) \in \Delta$

st $x = a\eta$,

$\alpha = r\tau$ and $\beta = \eta\tau$ for some $\tau \in \Gamma^*$.



$\vdash_P$



$(p, x, \alpha) \vdash_p^* (q, y, \beta)$ if $(p, x, \alpha) = (q, y, \beta)$,

$\exists (p_0, x_0, \alpha_0), \dots, (p_n, x_n, \alpha_n)$

s.t. $(p, x, \alpha) = (p_0, x_0, \alpha_0)$

$(p_{i-1}, x_{i-1}, \alpha_{i-1}) \vdash_p (p_i, x_i, \alpha_i) \forall i \in [1, n]$

$(p_n, x_n, \alpha_n) = (q, y, \beta)$

We say P accept a string $w \in \Sigma^*$ if

$(s, w, e) \vdash_p^* (q, e, e)$ for some $q \in \bar{F}$

initial state input start with empty stack final state consume all the input symbols stack is empty

② ① ③

$L(P) = \{ w \in \Sigma^* : P \text{ accepts } w \}$

P accepts $L(P)$

Exercise.

Design a PDA to accept $L = \{ ww^R : w \in \{a, b\}^* \}$

Idea.

push left half onto stack . . .

Solution

$P = (K, \Sigma, \Gamma, \Delta, s, \bar{F})$

• $K = \{l, r\}$

• $\Sigma = \{a, b\}$

- $T = \{a, b\}$
- $S = \{ \}$
- $F = r$
- $\Delta = \{ (l, a, e), (l, a), (l, b, e), (l, b), (l, e, e), (r, e), (r, a, a), (r, e), (r, b, b), (r, e) \}$

Exercise

Design a PDA to accept $L = \{w \in \{0,1\}^* : \#0's = \#1's \text{ in } w\}$

Idea.

When reading 0 / how to test?

If stack is empty or has 0 at the top,
push 0

else

pop 1

When reading 1

"

"



$P = (K, \Sigma, T, \Delta, s, F)$

$K = \{s, q, f\}$

$\Sigma = \{0, 1\}$

$T = \{0, 1, \$\}$

$$\Delta = \{ (s, e, e), (q, \$), (q, 0, \$), (q, 0\$), (q, 0, 0), (q, 00), (q, 0, 1), (q, e), (q, 1, \$), (q, 1\$), (q, 1, 1), (q, 11), (q, 1, 0), (q, e), (q, e, \$), (f, e) \}$$

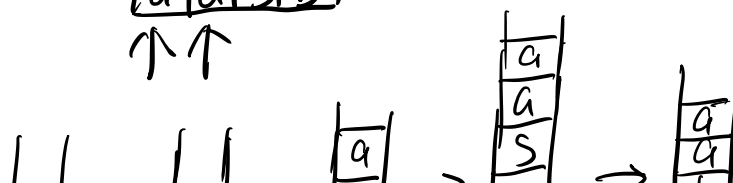
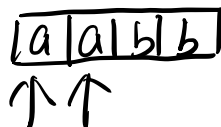
CFG $\Leftrightarrow$ PDA.

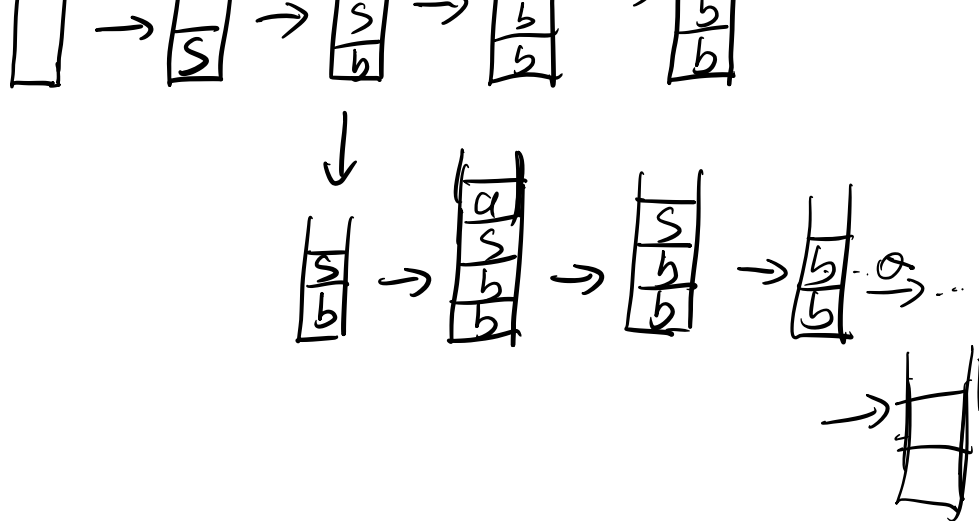
1. for each CFG G , there exists a PDA P s.t. $L(P) = L(G)$
2. for each PDA P , there exists a CFG G s.t. $L(G) = L(P)$.

Idea:

1. non-deterministically generates a string in stack using G
2. compare it to the input
3. accept input if they match

$$S \rightarrow e$$





Construction

$$G = (V, \Sigma, R, S)$$

$$P = (K, \Sigma, T, \Delta, p, F)$$

$$K = \{p, q\}$$

$$T = V$$

$$F = \{q\}$$

$$\Delta = \{ (p, e, e), (q, S), \\ (q, e, A), (q, u) \text{ for each } (A, u) \in R \\ (q, a, a), (q, e) \text{ for each } a \in \Sigma \\ \}$$

Example

$$S \rightarrow aSb$$

$$S \rightarrow e$$

$$K = \{p, q\} \quad \Sigma = \{a, b\}$$

$$T = \{S, a, b\} \quad F = \{q\}$$

$$\Delta = \{ (p, e, e), (q, S), \\ (q, e, S), (q, aSb), \\ (q, e, S), (q, e), \\ (q, a, a), (q, e), \\ (q, b, b), (q, e) \}$$

Prove $L(G) = L(P)$

$$\forall w \in \Sigma^*, S \Rightarrow^* w \text{ iff } \underbrace{(p, w, e) \vdash_P (q, w, S) \vdash_P^* (q, e, e)}$$

Claim

For any $w \in \Sigma^*$ and $\alpha \in ((V - \Sigma)V^*) \cup \{e\}$,

$$S \xRightarrow{*} w\alpha \text{ iff } (q, w, S) \vdash_P^* (q, e, \alpha)$$

$\alpha = e$

Proof:

$\Rightarrow$) by induction on # steps of the leftmost derivation

Base case: # step 0

$$S \xRightarrow{*} w\alpha \Rightarrow S = w\alpha$$

$$\Rightarrow w = e, \alpha = S$$

$$\Rightarrow (q, w, S) \vdash_P^* (q, e, \alpha)$$

Inductive hypotheses: when # steps = k, ... holds

Inductive step: when # steps = k+1

$$S \xRightarrow{*} u_1 \xRightarrow{*} u_2 \xRightarrow{*} \dots \xRightarrow{*} u_k \xRightarrow{*} w\alpha$$

Let A be the leftmost non-terminal in u_k

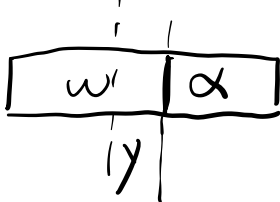
$$u_k = xA\beta \text{ where } x \in \Sigma^*, \beta \in V^*$$

$$w\alpha = x\gamma\beta \text{ for some } (A, \gamma) \in R$$



$$\exists y \in \Sigma^*, \text{ s.t.}$$

$$w = xy \text{ and } y\alpha = r\beta$$



$$((q, e, A), (q, r))$$

$$S \xRightarrow[k \text{ steps}]{L^*_{(XAB)} u_k} \Rightarrow (q, x, S) \vdash_p^* (q, e, A\beta) (\vdash_p (q, e, r\beta))$$

$$\Rightarrow (q, x, S) \vdash_p^* (q, e, r\beta)$$

$$\Rightarrow (q, xy, S) \vdash_p^* (q, y, r\beta)$$

$$\Rightarrow (q, w, S) \vdash_p^* (q, y, y\alpha) \vdash_p^* (q, e, \alpha)$$

□

⇐) by induction on # type-2 transition

Base case: # type-2 = 0

$$(q, w, S) \vdash_p^* (q, e, \alpha) \Rightarrow \begin{matrix} w=e \\ S=\alpha \end{matrix}$$

$$\Rightarrow S \xRightarrow{*} \frac{w\alpha}{S}$$

inductive hyp. when # type-2 = k, " " holds

inductive step: when # type-2 = k+1

let $((q, e, A), (q, r))$ be the last type-2 transition.

$$\exists y \in \Sigma^*, \exists \beta \in V^*$$

$$(q, w, S) \vdash_p^* (q, y, A\beta) \vdash_p (q, y, r\beta) \vdash_p^* (q, e, \alpha)$$

⏟

k type-2 transition

⏟

last # type-2 transition

⏟

type-3 transition

w

$$\boxed{x|y}$$

①

②

③

$$\textcircled{1} \Rightarrow \exists x \in \Sigma^*, w = xy$$

$$(q, xy, S) \vdash_p^* (q, y, A\beta)$$

$$\Rightarrow (q, x, S) \vdash_p^* (q, e, A\beta)$$

$$\Rightarrow S \xRightarrow{*} xA\beta$$

$$\textcircled{2} \Rightarrow (A, r) \in R \Rightarrow$$

$$x r \beta$$

$$\textcircled{3} \Rightarrow \exists \alpha, \gamma \alpha = r \beta$$

$$S \xRightarrow{*} \underline{x\gamma\alpha}.$$

$$w\alpha$$

o

o